

Experimental evaluation of IEEE 802.11ax - low latency and high reliability with Wi-Fi 6?

Marie-Theres Suer

Bosch Research

Hildesheim, Germany

Marie-Theres.Suer@de.bosch.com

Prince Jose

Hochschule Bremen

Bremen, Germany

pjose@stud.hs-bremen.de

Hugues Tchouankem

Bosch Research

Hildesheim, Germany

HuguesNarcisse.Tchouankem@de.bosch.com

Abstract—IEEE 802.11ax, also named Wi-Fi 6, introduces significant enhancements to the channel access mechanism. The introduction of orthogonal frequency-division multiple access (OFDMA) and the more centralized channel access coordinated by the AP can potentially improve the latency and reliability performance of Wi-Fi 6 compared to previous Wi-Fi generations. This would make Wi-Fi systems suitable also for more demanding applications, e.g., in the industrial domain, which require low communication latency with high reliability. However, only few works exist that evaluate these new features with regard to such applications. This work presents a comprehensive measurement study on the latency and reliability performance of Wi-Fi 6 and compares it to performance of Wi-Fi 5. The baseline scenario measurements reveal a significant latency reduction of Wi-Fi 6 compared to Wi-Fi 5 especially for larger number of stations and especially in uplink direction. Since Wi-Fi 6 operates in unlicensed bands like its predecessors, the influence of interfering networks on the communication performance is an important aspect. Our evaluation shows that interfering traffic can significantly increase the latency and packet loss experienced by Wi-Fi 6 for smaller operation bandwidths, while for larger bandwidth of 80 MHz and resulting lower relative network load Wi-Fi 6 can still provide 99-percentile latency below 10 ms in the evaluated scenario. Overall, the achieved results with Wi-Fi 6 are promising, but further evaluations especially on the effect of different OFDMA scheduling algorithms and configurations is needed to evaluate its potential for demanding applications.

Index Terms—802.11ax, URLLC, reliable low-latency communications, Wi-Fi

I. INTRODUCTION

The latency and reliability performance of wireless communication systems is of major importance for emerging applications such as mobile robotics. The IEEE real-time applications technical interest group (RTA TIG) specified a latency requirement of 10-50 ms that needs to be achieved with a reliability of 99%-99.9% for mobile robots [1]. Former Wi-Fi versions reach low latency for few users, but experience high latency for higher number of stations even in scenarios without interfering networks [2]. The 802.11ax amendment, also named Wi-Fi 6, introduces among other features an OFDMA-based channel access scheme that is centrally coordinated by the access point (AP) [3]. The AP still needs to gain channel access according to the carrier sense multiple access/collision avoidance (CSMA/CA) scheme as it operates in unlicensed

spectrum. However, once it gained access to the channel the AP coordinates the transmissions of stations in both uplink (UL) and downlink (DL) direction and due to the OFDMA-feature the AP can sub-divide the channel frequency resources to different clients. The different channel access mechanisms in Wi-Fi 5 and Wi-Fi 6 are illustrated in Figure 1 for the UL direction. For Wi-Fi 5 each station individually uses the CSMA/CA based channel access method, while for Wi-Fi 6 the AP gains channel access and then sends a trigger frame to the stations which then start their UL transmission in their respective scheduled resources. The novel channel access scheme in Wi-Fi 6 promises improvements in terms of communication latency and reliability, but needs to be thoroughly investigated in different scenarios.

In [4] the influence of the OFDMA-based channel access scheme on latency and throughput is evaluated in ns-3 simulations. For the latency evaluation only a single scenario with all devices in similar signal conditions is considered. The experimental evaluations of Wi-Fi 6 in [5] and [6] do not consider OFDMA, but only evaluate the impact of the higher modulation coding scheme (MCS) available in Wi-Fi 6. In [7] the performance of DL OFDMA was evaluated with three clients generating iPerf traffic with traffic profiles for home applications for AP reference boards from two different vendors. The focus is on the throughput performance and on the percentage of packets for which OFDMA is applied which is found to vary strongly between the two vendors. The authors of [8] compare the throughput performance of Wi-Fi 5 and Wi-Fi 6 in a residential scenario with different load and different signal strengths. They find that Wi-Fi 6 can especially increase the throughput in scenarios with good signal strength.

To the best of our knowledge, this work is the first that experimentally evaluates the Wi-Fi 6 latency and reliability performance for different network loads. Additionally, we evaluate the impact of background traffic applications as well as interfering networks and the impact of stations operating with different signal strength on the performance of Wi-Fi 6. Our study applies cyclic traffic profiles as utilized by many industrial applications and evaluates both UL and DL performance of the OFDMA-based channel access scheme with COTS hardware. The rest of this paper is organised as follows: Section II introduces the testbed setup, measurement tools,

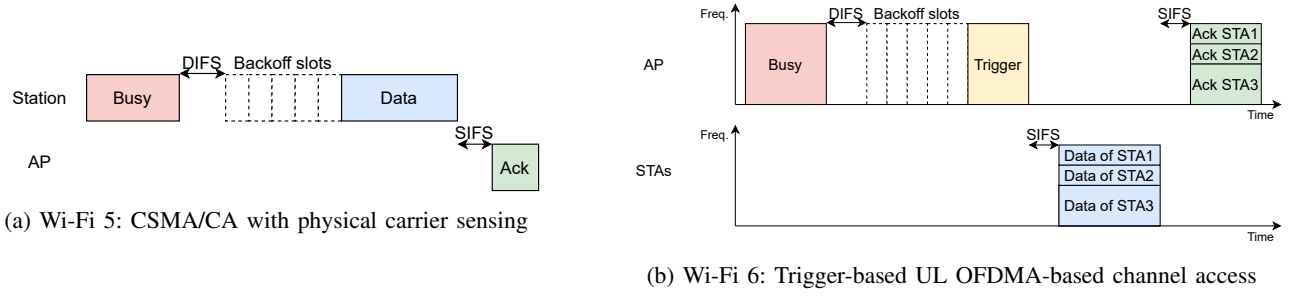


Fig. 1: Illustration of the channel access mechanisms in Wi-Fi 5 and Wi-Fi 6 in UL direction. Adapted from [3].

equipment and scenarios applied in our study. In Section III the results for three different scenarios are presented and discussed: a baseline scenario, two scenarios with background traffic interference and a scenario where one station operates with low signal strength, while the other stations remain in good signal conditions. Our findings are concluded in Section IV and courses for future work are highlighted.

II. MEASUREMENT SETUP

For the experimental evaluation of Wi-Fi 5 and Wi-Fi 6 the testbed illustrated in Figure 2 was utilized. The Cisco C9120 AP is connected via Ethernet to a server which serves as one communication endpoint. On the client side fitlet2 miniPCs were utilized and equipped with Intel ax200 Wi-Fi cards. The measurements were conducted in the 5 GHz band with no other Wi-Fi networks present in the utilized channels, so no uncontrolled interference from other networks affected the measurements. Server and fitlets were time-synchronized with precision time protocol (PTP) over a separate Ethernet connection to enable one-way delay measurements. As measurement tool RUDE & CRUDE was utilized which can generate cyclic traffic with configurable inter-packet-gap (IPG) and payload size. For the measurement results presented in this paper a payload size of 1400 Byte and an IPG of 1 ms were applied. This corresponds to the conventional industrial control use case as defined by 3GPP [9]. As evaluation metric a combined latency and reliability metric was applied where lost packets are assigned a latency of infinity. This metric is useful for industrial applications as it makes no difference for most applications if the packet is lost or if the packet delay is higher than the specific latency deadline.

For additional analysis, a sniffer node running tshark was placed close to the clients. Also, the statistics gathered by the AP, e.g., about the resource unit (RU) allocation in Wi-Fi 6, were automatically saved to files for deeper analysis. The clients are placed in line-of-sight to the AP and report a received signal strength of about -50 dBm. For the scenarios with background traffic two extra clients ran iPerf with target data rate of 50 Mbps. In the first scenario these were using the same Wi-Fi AP as the fitlets running RUDE & CRUDE. In the second background traffic scenario an additional AP which operates on the same channel as the main AP is utilized for the background traffic stations. This AP applies Wi-Fi 5. In

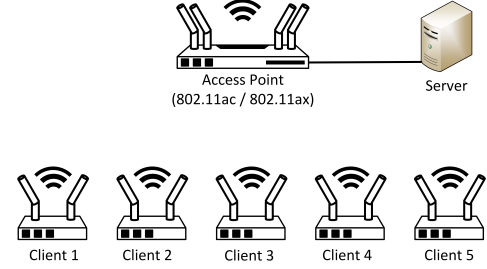


Fig. 2: Testbed architecture.

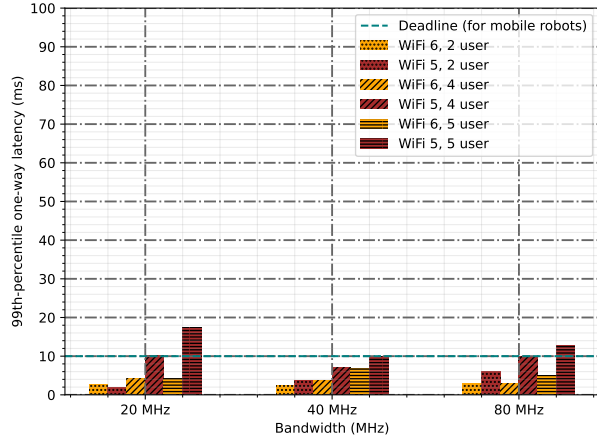
a third scenario attenuators were attached to the antennas of station one resulting in a lower RSSI of -80 dBm for this station, while the other stations remained in the good signal conditions with -50 dBm signal strength.

III. RESULTS AND DISCUSSION

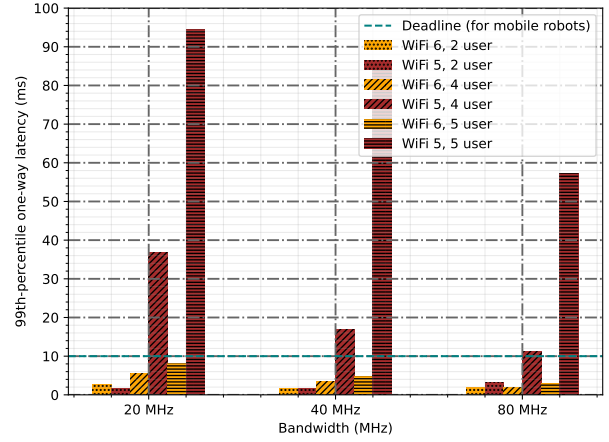
In the following the performance of 802.11ax or Wi-Fi 6 is compared to the performance of 802.11ac or Wi-Fi 5 in different scenarios. For each scenario different number of active clients and different operation bandwidths are considered. The latency is evaluated in terms of the 99th-percentile latency, which means that 99% of the transmitted packets achieve a latency of this value or lower. Lost packets are also considered in this metric and are counted as packets with latency of infinity.

A. Baseline scenario

For the baseline scenario, all nodes operate with strong signal strength of about -50 dBm and no other networks or background applications are present. Figure 3 presents the 99th-percentile latency for different numbers of active stations and for different operation bandwidths for UL and DL. The 99th-percentile latency also includes lost packets and is evaluated over all active stations. For both UL and DL an increase in latency can be observed with increasing number of users. Wi-Fi 6 can significantly reduce the latency compared to Wi-Fi 5 especially for larger number of users and for larger bandwidths. The latency reduction with Wi-Fi 6 is higher in UL than in DL. The Wi-Fi 6 AP can distribute the frequency resources to multiple clients, while in Wi-Fi 5 only one client can use the whole channel width at a time. This is the main reason for the latency reduction for larger number of clients

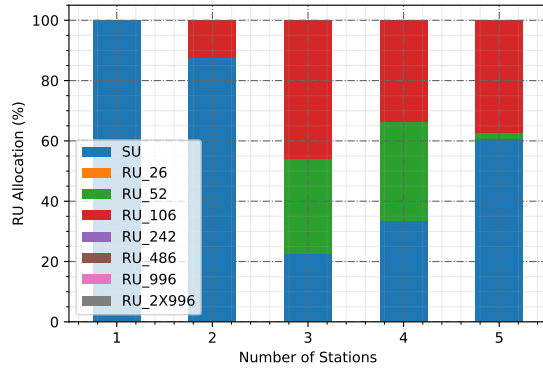


(a) DL

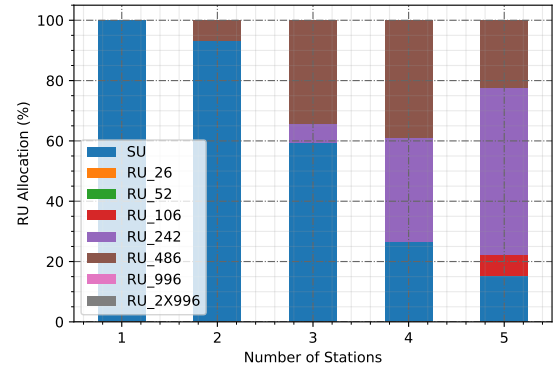


(b) UL

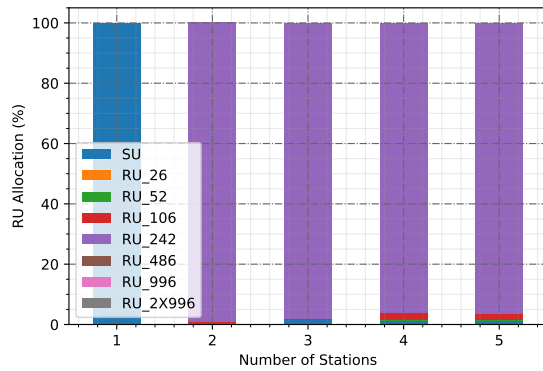
Fig. 3: 99th-percentile latency for the baseline scenario for different channel widths comparing Wi-Fi 5 and Wi-Fi 6 performance for different number of users in DL and UL. Lost packets are considered with a latency of infinity.



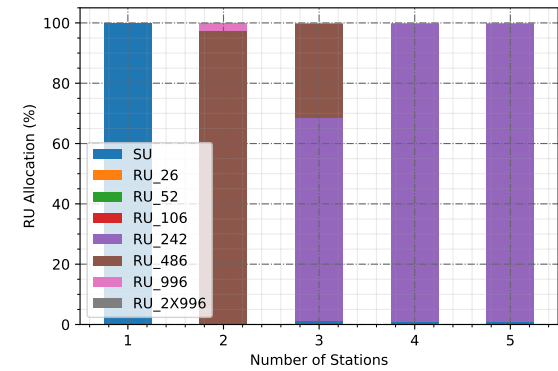
(a) DL 20 MHz



(b) DL 80 MHz



(c) UL 20 MHz



(d) UL 80 MHz

Fig. 4: Statistics for the allocation of resource units (RUs) in Wi-Fi 6 for the baseline scenario with 20 MHz and 80 MHz channel bandwidth in UL and DL for different number of active stations.

	20 MHz UL	20 MHz DL	40 MHz UL	40 MHz DL	80 MHz UL	80 MHz DL
1 Client	8.00	8.00	9.00	7.11	8.95	7.15
5 Clients	8.00	5.88	8.99	7.63	8.91	5.51

TABLE I: Mean MCS values for Wi-Fi 5 in the baseline scenario.

in DL. In UL the clients need to apply the distributed channel access scheme CSMA/CA for Wi-Fi 5, while for Wi-Fi 6 only the AP applies CSMA/CA and once it gains access to the channel it centrally coordinates the access of the clients. This results in a drastically reduced latency especially for larger number of clients as the centrally managed scheme is more efficient. In UL a clear trend is observable that for increasing operation bandwidth the latency for both Wi-Fi 5 and Wi-Fi 6 is decreasing due to the higher capacity. In DL this trend is also observable for Wi-Fi 6, while for Wi-Fi 5 the trend is not as clear and the latency is even slightly increasing for 80 MHz bandwidth compared to 40 MHz bandwidth for four and five active stations. The mean MCS values for Wi-Fi 5 for different bandwidths for a single and for 5 users are displayed in Table I. A lower applied MCS can be observed in DL with growing bandwidth, while for UL the MCS remains relatively constant. A reason for this might be different rate adaptation algorithms implemented in the client device for UL and implemented in the AP for DL transmissions. The negative impact of the larger operation bandwidth on the signal-to-noise ratio (SNR) is potentially smaller in UL compared to DL due to expected higher quality transceivers utilized in the AP compared to the clients. This could lead to the increased latency values observed for 80 MHz compared to 40 MHz operation bandwidth for five active stations with Wi-Fi 5.

To gain a deeper understanding of the OFDMA operation the statistics of the applied RU allocation were collected in the AP and are presented in Figure 4. The plot displays the percentage of single user (SU) and different RU size transmissions utilized for different number of users. For 20 MHz operation bandwidth the largest usable RU is "RU_242", so a RU with 224 subcarriers, and for 80 MHz "RU_996", so a RU with 996 subcarriers. In DL the number of SU transmissions where the whole channel is assigned to a single user generally decreases with increasing number of users. The size of the assigned RUs also decreases with growing number of users such that more stations can be scheduled in the same time frame. This trend can also be observed in UL. In UL hardly any SU transmissions are utilized when multiple stations are present. The reason for the lower utilization of SU transmission in UL compared to DL is likely that the positive effect of scheduled transmissions is larger in UL where the stations would otherwise need to compete for the channel with CSMA/CA. In general, the usage of OFDMA transmissions depends on the number of active clients, the communication direction and the operation bandwidth: higher OFDMA usage can be observed for larger number of clients, in UL more OFDMA transmissions are used than in DL and higher OFDMA usage can be observed for increasing

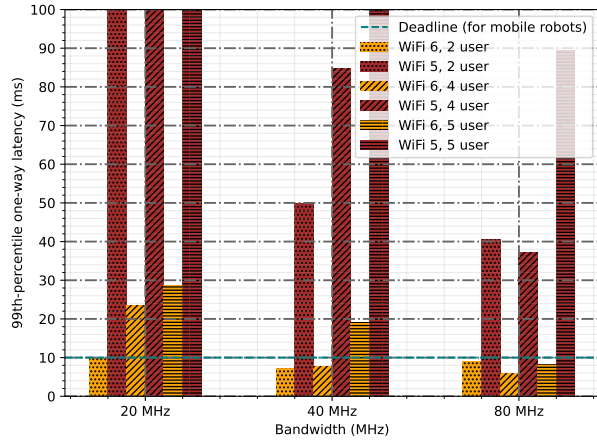
operation bandwidth. The scheduling algorithm that activates OFDMA and distributes the RUs to the stations depends on the proprietary implementation in the AP and thus varies for different vendors and configurations.

B. Background traffic scenarios

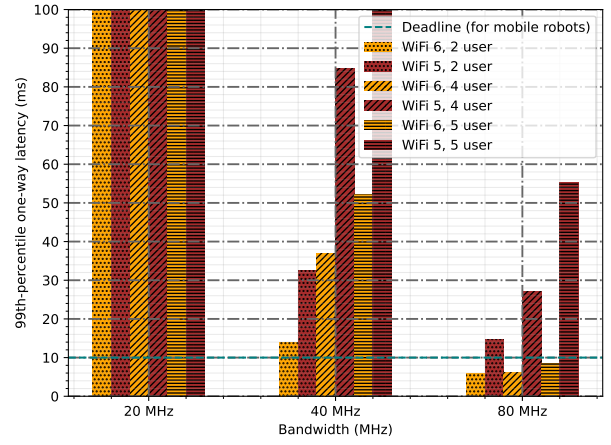
For the background traffic scenarios two additional clients are utilized each creating 50 Mbps traffic with iPerf in UL direction. In the first background traffic scenario these two clients are connected to the same network as the nodes with latency critical traffic running RUDE & CRUDE. In the second scenario the nodes are connected to a separate Wi-Fi 5 AP that was operating on the same channel and thus was interfering with the network AP the latency critical traffic.

Figure 5a shows the 99th-percentile latency for the scenario with background traffic in the same network in UL direction. The 99th-percentile latency is evaluated for the stations running the latency-critical traffic with RUDE & CRUDE. Compared to the scenario without background traffic, cf. Figure 3b, higher latency can be observed for both Wi-Fi 5 and Wi-Fi 6. However, the latency achieved with Wi-Fi 6 is much lower compared to Wi-Fi 5. The background traffic with Wi-Fi 6 can also be centrally scheduled by the AP utilizing the OFDMA-based channel access scheme and therefore the resource sharing is much more efficient compared to Wi-Fi 5 where the clients gain access in a distributed manner utilizing the CSMA/CA scheme. This is also reflected in the throughput achieved by the background traffic applications: for 20 MHz operation bandwidth the background traffic stations achieve the full 50 Mbps for Wi-Fi 6 and only 44 Mbps for Wi-Fi 5.

The 99th-percentile latency for the scenario with background traffic using a separate AP that interferes with the AP bearing the latency critical traffic is shown in Figure 5b for the UL direction. For 20 MHz operation bandwidth the Wi-Fi 6 latency is larger than the Wi-Fi 5 latency for 5 active stations: Wi-Fi 5 experiences a 99th-percentile latency of about 250 ms, while for Wi-Fi 6 a value of about 470 ms is observed. For larger bandwidths the Wi-Fi 6 latency is significantly lower than the Wi-Fi 5 latency. The reason for this is that in UL for Wi-Fi 5 each station itself is competing for the channel access, while for Wi-Fi 6 only the AP is competing for the channel access and once it gains access it schedules the stations UL transmissions. The two background traffic stations are using Wi-Fi 5 and are therefore also competing for the channel access individually. Therefore, for Wi-Fi 5 the stations are more likely to gain channel access as multiple nodes are trying compared to Wi-Fi 6 where only the AP tries to gain access to the channel. This is why for the scenario with five active stations and an operation bandwidth

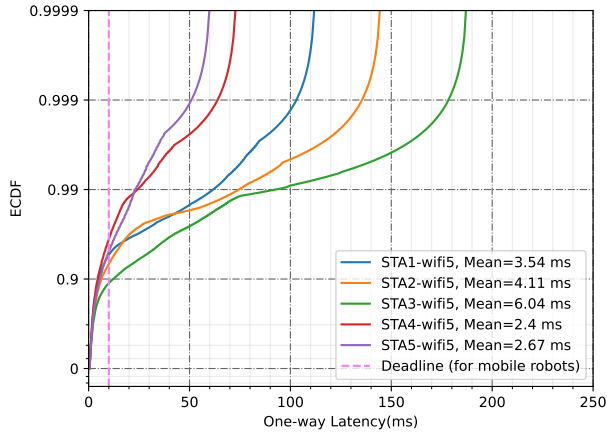


(a) Background traffic in same network

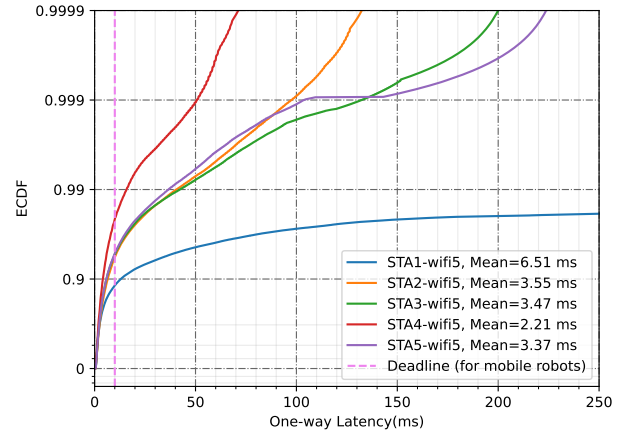


(b) Background traffic in interfering network

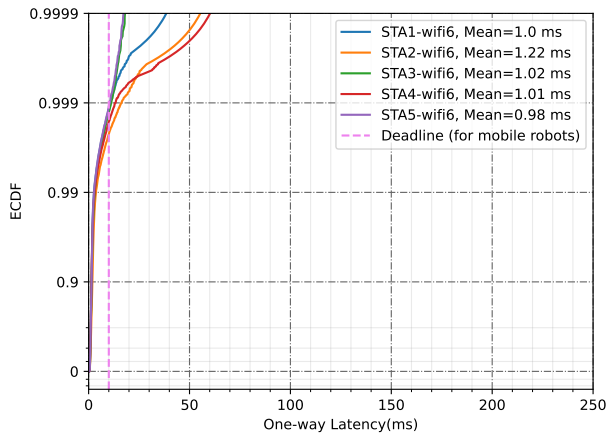
Fig. 5: 99th-percentile latency for the scenarios with external background traffic interference for different channel widths comparing Wi-Fi 5 and Wi-Fi 6 performance for different number of active stations in UL direction. Lost packets are considered with a latency of infinity.



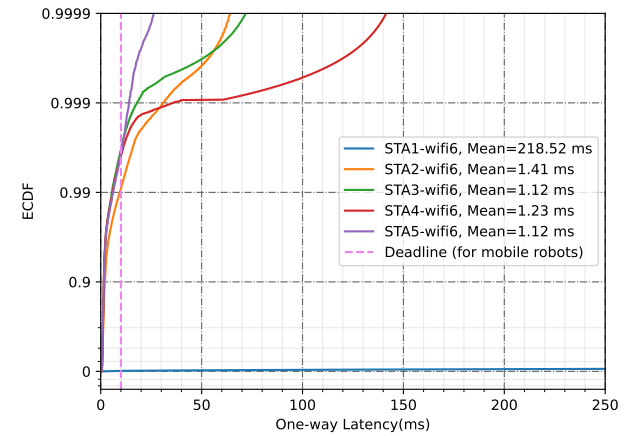
(a) Wi-Fi 5 baseline



(b) Wi-Fi 5 with attenuators at station one



(c) Wi-Fi 6 baseline



(d) Wi-Fi 6 with attenuators at station one

Fig. 6: Empirical cumulative distribution function of the combined latency and reliability metric for each client with 5 active stations for a bandwidth of 80 MHz in UL. Lost packets are considered with a latency of infinity.

of 20 MHz Wi-Fi 5 outperforms Wi-Fi 6. For 40 and 80 MHz channel width on the other hand Wi-Fi 6 outperforms Wi-Fi 5. This is because Wi-Fi 6, when it accesses the channel, can use the transmit opportunities more efficiently by sharing the channel bandwidth among the different stations. Nevertheless, the results show that Wi-Fi 6 also suffers high latency in scenarios with interfering networks and high relative network load.

C. Scenario with nodes operating in different SNRs

While in the previous scenarios all nodes were operating with strong signal strength of -50 dBm, in this scenario attenuators were added to the antennas of station one and this station was therefore experiencing a signal strength of -80 dBm. Figure 6 displays the empirical cumulative distribution function (ECDF) for each station in a scenario with 5 stations for a bandwidth of 80 MHz in UL. The baseline scenario is compared to the scenario with attenuators at station one. For Wi-Fi 5 the performance of all stations degrades significantly compared to the baseline scenario. The station with the attenuators, i.e. station one, performs worse than the other stations and experiences a higher mean latency and higher packet loss. The reason for this is that the nodes compete for the channel with CSMA/CA in Wi-Fi 5 and all stations have similar probability to gain access to the channel. The attenuators cause station one to transmit with lower MCS and thus occupy the channel longer which leads to higher latencies also for the other stations. For Wi-Fi 6 on the other hand the performance of the stations without attenuators stays relative constant compared to the baseline scenario with only small increase in mean and 99th-percentile latency. The performance of station one, the station with lower signal strength, on the other hand degrades drastically. In Wi-Fi 6 the channel access is centrally managed by the AP. For the evaluated AP and scheduling algorithm nodes with low SNR and therefore lower MCS and resulting lower transmission rate seem to be scheduled with much lower priority. This leads to very high latency for the station with attenuators, while the performance of the other stations is not drastically affected. This behaviour strongly depends on the scheduling algorithm employed in the AP and might therefore vary for devices from different vendors or with different configurations. Depending on the application requirements this behaviour could be beneficial as stations with good signal conditions seem to have stable low latency, but could also be problematic for applications where it is important that all stations achieve similar performance independent of their SNR.

IV. CONCLUSIONS

IEEE 802.11ax, also named Wi-Fi 6, introduces OFDMA and a more centralized channel access which could potentially make Wi-Fi systems more suitable for applications demanding low communication latency with high reliability. In this work a measurement study has been performed comparing the latency and reliability performance of Wi-Fi 5 and Wi-Fi 6. A significant latency reduction can be observed with Wi-Fi 6

compared to Wi-Fi 5 for increasing number of stations and especially in UL direction. The usage of OFDMA in Wi-Fi 6 was found to strongly depend on the number of active stations, the communication direction and the operation bandwidth. To the best of our knowledge this is the first study that evaluates the impact of interfering background traffic applications on the Wi-Fi 6 latency and reliability performance. Background traffic applied on the same network was found to be handled well by Wi-Fi 6 and not significantly impact the performance of the latency-critical traffic. For scenarios with background traffic applied on a separate network in the same channel, the Wi-Fi 6 performance degraded for smaller bandwidth and high network load scenarios, while for larger bandwidths it could still provide a stable 99th-percentile latency below 10ms.

Overall, the measurement results highlight the improved latency and reliability performance of Wi-Fi 6 compared to previous Wi-Fi generations. The requirements of the mobile robotics use case of a 10 ms latency deadline being met by 99% of the packets could be achieved in all considered scenarios provided that a bandwidth of 80 MHz was applied. The performance of Wi-Fi 6 is strongly dependent on the implementation and configuration of the OFDMA scheduling algorithm in the AP. Therefore, evaluations with AP chipsets from different vendors or simulative studies where the OFDMA scheduler can be influenced and optimized for specific application requirements are relevant directions for further studies. Also, scenarios with larger number of clients, with different traffic models and varying interfering traffic could be valuable to understand the limitations of the OFDMA-based channel access scheme.

REFERENCES

- [1] *IEEE 802.11 Real Time Applications TIG Report; doc.:IEEE 802.11-18/2009r2*, IEEE Std., November 2018.
- [2] M.-T. Suer, C. Thein, H. Tchouankem, and L. C. Wolf, "Reliability and Latency Performance of Multi-Connectivity Scheduling Schemes in Multi-User Scenarios," in *Proc. of the 2020 32nd International Teletraffic Congress (ITC 32)*, Osaka, Japan, Sep. 2020.
- [3] *Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications; Amendment 1: Enhancements for High-Efficiency WLAN*, IEEE Std., May 2021.
- [4] S. Avallone, P. Imputato, G. Redieteb, C. Ghosh, and S. Roy, "Will OFDMA Improve the Performance of 802.11 Wifi Networks?" *IEEE Wireless Communications*, vol. 28, no. 3, pp. 100–107, Jun. 2021.
- [5] A. F. Rochim, B. Harijadi, Y. P. Purbanugraha, S. Fuad, and K. A. Nugroho, "Performance comparison of wireless protocol IEEE 802.11ax vs 802.11ac," in *Proc. of the 2020 International Conference on Smart Technology and Applications (ICoSTA)*, Surabaya, Indonesia, Feb. 2020.
- [6] J. Sánchez-Mahecha, S. Céspedes, and J. Bustos-Jiménez, "QoS Evaluation of the Future High-Efficiency IEEE 802.11ax WLAN Standard," in *Proc. of the 2018 IEEE Colombian Conference on Communications and Computing (COLCOM)*, Medellín, Colombia, May 2018.
- [7] D. Weller, R. Mensenkamp, A. van der Vegt, J.-W. van Bloem, and C. de Laat, "Wi-Fi 6 performance measurements of 1024-QAM and DL OFDMA," in *Proc. of the 2020 IEEE International Conference on Communications (ICC)*, Dublin, Ireland, Jun. 2020.
- [8] J. Sandoval and S. Céspedes, "Performance Evaluation of IEEE 802.11ax for Residential Networks," in *Proc. of the 2021 IEEE Latin-American Conference on Communications (LATINCOM)*, Santo Domingo, Dominican Republic, Nov. 2021.
- [9] *Technical Specification Group Services and System Aspects; Feasibility Study on New Services and Markets Technology Enablers for Critical Communications; Stage 1 (Release 14); 3GPP TR 22.862 V14.1.0*, 3rd Generation Partnership Project Std., October 2016.